Program 6: sliding window protocol

Aim : Develop a simple data link layer that performs the flow control using the sliding window protocol, and loss recovery using the Go-Back-N mechanism.

Description:

Sliding Window Protocol : Sliding Window Protocol is a flow-control and error-control technique that allows the sender to transmit multiple frames before receiving acknowledgements. The sender maintains a window of frames that can be sent, and as ACKs are received, the window **slides forward** to allow new frames. It improves channel utilization and transmission efficiency compared with Stop-and-Wait.

Go-Back-N Protocol : Go-Back-N is a sliding-window ARQ error-control protocol in which multiple frames can be transmitted without waiting for individual ACKs. If a frame is lost or corrupted, the receiver discards that frame and subsequent out-of-order frames, and the sender retransmits the erroneous frame and all following frames in the window. It provides reliable transmission but may cause unnecessary retransmissions.

Program:

```
#include <stdio.h>

#define MAX_FRAMES 50

int main()

{

   int n, windowSize;

   int base = 0;

   int nextFrame = 0;

   int ack[MAX_FRAMES] = {0};

   int lostFrame;

   printf("Enter number of frames: ");

   scanf("%d", &n);

   printf("Enter window size: ");

   scanf("%d", &windowSize);

   printf("Enter the frame number to simulate loss (-1 for no loss): ");

   scanf("%d", &lostFrame);

  printf("\n--- Go-Back-N Sliding Window Protocol ---\n\n");

   while (base < n)
```

```
{
    /* Send frames within the window */
    while (nextFrame < base + windowSize && nextFrame < n)
    {
        printf("Sender: Sending Frame %d\n", nextFrame);
        nextFrame++;
    }
    /* Receiver processes frames */
    int i;
    for (i = base; i < nextFrame; i++)
    {
        if (i == lostFrame && ack[i] == 0)
        {
            printf("Receiver: Frame %d LOST!\n", i);
            printf("Receiver: Discarding Frame %d and later frames\n", i);
            printf("Sender: Timeout for Frame %d\n", i);
            printf("Sender: Go-Back-N retransmission starts\n\n");
            /* Retransmit from lost frame */
            nextFrame = i;
            lostFrame = -1;  // Loss occurs only once
            break;
        }
        else if (ack[i] == 0)
        {
            printf("Receiver: Frame %d received correctly\n", i);
            printf("Receiver: ACK %d sent\n", i);
            ack[i] = 1;
            base = i + 1;
        }
    }
```

```
        printf("\n");
    }
    printf("All frames received successfully.\n");
    printf("Data transmission completed using Go-Back-N.\n");
    return 0;
}
```

Output:

```
--- Go-Back-N Sliding Window Protocol ---

Sender: Sending Frame 0
Sender: Sending Frame 1
Sender: Sending Frame 2

Receiver: Frame 0 received correctly
Receiver: ACK 0 sent
Receiver: Frame 1 LOST!
Receiver: Discarding Frame 1 and later frames
Sender: Timeout for Frame 1
Sender: Go-Back-N retransmission starts

Sender: Sending Frame 1
Sender: Sending Frame 2
Sender: Sending Frame 3

Receiver: Frame 1 received correctly
Receiver: ACK 1 sent
Receiver: Frame 2 received correctly
Receiver: ACK 2 sent
Receiver: Frame 3 received correctly
Receiver: ACK 3 sent
```

Viva Question:

1. **What is Stop-and-Wait protocol?**
   Answer: The sender sends one frame at a time and waits for an acknowledgement before sending the next frame. It has low efficiency, especially when the propagation delay is large, because the sender remains idle while waiting for an ACK.

2. **What is the purpose of ACK?**
   Answer: ACK confirms that a frame has been received correctly.
3. **What happens if an ACK is lost?**
   Answer: The sender's timer expires and it retransmits the frame.
4. **What is a noisy channel?**
   Answer: A noisy channel is a communication channel where transmitted frames or acknowledgements may be **lost or corrupted**.
5. **How does Stop-and-Wait handle a noisy channel?**
   Answer: It uses ACKs, sequence numbers, timers, and retransmission**.**
6. **What is ARQ?**
   Answer: ARQ (**Automatic Repeat reQuest**.) is an error-control technique in which the receiver acknowledges correctly received frames and the sender retransmits frames that are lost or corrupted.
7. **What is a Sliding Window Protocol?**
   Answer: It allows the sender to transmit multiple frames before receiving acknowledgements, up to the size of the transmission window.
8. **What is Go-Back-N ARQ?**
   Answer: Go-Back-N is a sliding-window ARQ technique where, if a frame is lost or corrupted, the sender retransmits that frame and all subsequent frames in the current window.
9. **What type of acknowledgement is commonly used in Go-Back-N?**
   Answer: Cumulative ACKs are commonly used. A cumulative ACK confirms that all frames up to a particular point have been received correctly.
10. **What is Selective Repeat ARQ?**
    Answer: Selective Repeat is a sliding-window protocol that retransmits only the lost or corrupted frames.
11. **Which is more efficient in a noisy channel?**
    Answer: Selective Repeat, because it retransmits only the frames that were lost or corrupted.
12. **What is piggybacking?**
    Answer: Piggybacking is a technique where an ACK is attached to an outgoing data frame instead of sending a separate ACK frame. It reduces the number of control frames, saving bandwidth and transmission overhead**.** It is especially useful in bidirectional communication, improving channel efficiency and reducing delay.

13. What is Pipelining ?

Answer: Pipelining is a technique in which a sender can send multiple frames before receiving an acknowledgment (ACK) for the previous frame. Pipelining improves transmission efficiency, especially when the bandwidth-delay product (BDP) is large